

Fine-Grained Accident Detection: Database and Algorithm

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Abstract—This paper presents a novel fine-grained task for traffic accident analysis. Accident detection in surveillance or dashcam videos is a common task in the field of traffic accident analysis by using videos. However, common accident detection does not analyze the specific particulars of the accident, only identifies the accident's existence or occurrence time in a video. In this paper, we define the novel fine-grained accident detection task which contains fine-grained accident classification, temporal-spatial occurrence region localization, and accident severity estimation. A transformer-based framework combining the RGB and optical flow information of videos is proposed for fine-grained accident detection. Additionally, we introduce a challenging Fine-grained Accident Detection (FAD) database that covers multiple tasks in surveillance videos which places more emphasis on the overall perspective. Experimental results demonstrate that our model could effectively extract the video features for multiple tasks, indicating that current traffic accident analysis has limitations in dealing with the FAD task and that further research is indeed needed.

Index Terms—Traffic accidents, accident classification, accident detection, severity estimation.

I. INTRODUCTION

THE analysis of traffic accidents with videos has developed tremendously in recent years [1], [2], [3], [4]. The number of traffic accidents is inevitably increasing due to the rapid expansion of the city scale and the rapid increase of vehicle amount. Traffic congestion and traffic accidents have become the biggest headaches for traffic management departments. Much more precise accident detection is required for smart city construction. Effective detection of traffic accidents can reduce the loss of property and casualties caused by delayed rescue, so it may not be feasible to use human resources to

monitor every single screen and make judgments. With the popularization of camera equipment, it is possible to automatically detect and analyze traffic accidents using computer vision technology with monitoring data.

Two kinds of videos are usually used for traffic accident analysis: the dashcam inside the vehicle's interior and the stationary surveillance camera on the street. The dashcam can provide the first view of the accident, showing the accident details from the perspective of the participants or the witnesses in the neighborhood [1], [4]. The dashcam is effective for individual applications such as self-driving cars. It could be useful for understanding the scene situation and detecting or predicting the emergencies. However, the dashcam can only display a limited view of the scene, and may not be able to fully capture the overall perspective of the accident. Stationary surveillance cameras are always placed at intersections or on either side of roads, which facilitate the official traffic management department to observe and judge the traffic conditions of a certain location [2], [5]. They can offer a much higher and wider view of the scene, recording the whole process of the accident happening and being suitable for estimating overall accident severity and the affected area. In this paper, we focus on stationary surveillance cameras to analyze the fine-grained attributes of traffic accidents.

Traffic accidents are complex and varied, with various occurrence patterns. However, general traffic accident analysis is only considered as detecting the existence, as shown in Table I. The analysis in terms of the kind and context of the accident has not been carefully considered. Most methods, challenges, and databases treat traffic accidents as part of anomaly detection and make a 0/1 binary classification [1], [6], i.e., whether the traffic accident occurred. However, the severity or situation of the impact of different kinds of traffic accidents also varies, which should be dealt with using different measures. Therefore, we redefine the classification of the traffic accidents, extending the 0/1 binary classification accident detection task to multiple fine-grained classification tasks.

Meanwhile, accident analysis usually focuses on temporal level analysis, which means detecting the moment the traffic accident happened and ended. However, the spatial analysis of traffic accidents is still lacking, which remains at the level of object detection in the scene [2], [4]. Therefore, we focus on the spatial analysis of the area where traffic accidents occurred and define the event-level region detection of traffic accidents.

Traffic accident detection should not only be limited to the analysis of video at the temporal and spatial level but also

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TABLE I

SUMMARY OF EXISTING RELATED DATABASES. T-DETECTION INDICATES TEMPORAL DETECTION. C/TS/S-DETECTION INDICATES CLASSIFICATION, TEMPORAL-SPATIAL DETECTION, AND ACCIDENT SEVERITY ESTIMATION RESPECTIVELY. 0/1 INDICATES BINARY CLASSIFICATION. 0/26 INDICATES NORMAL CLASS AND 26 ACCIDENT CLASSES

Databases	Attributes									
	Time	Space	Classes	Videos	Accident	Weather	Severity	Tasks	View	Year
DAD [1]	✓	objects	0/1	1,730	620			Anticipation	Dashcam	2016
A3D [9]	✓		0/1	1,500	1,500			T-Detection	Dashcam	2019
DADA [10]	✓		62	2,000	2,000	✓		Attention	Dashcam	2019
CCD [4]	✓	objects	0/1	4,500	1,500	✓		Anticipation	Dashcam	2020
Iowa DOT [6]	✓		0/1	200	50			T-Detection	Surveillance	2020
CADP [2]	✓	objects	0/1	1,416	1,416			Anticipation	Surveillance	2018
IITH [5]	✓		0/1	7	7			T-Detection	Surveillance	2020
UCF Crime [11]	✓		0/1	1,900	150			T-Detection	Surveillance	2018
FAD	✓	events	0/26	3,996	2,393	✓	✓	C/TS/S-Detection	Surveillance	2022

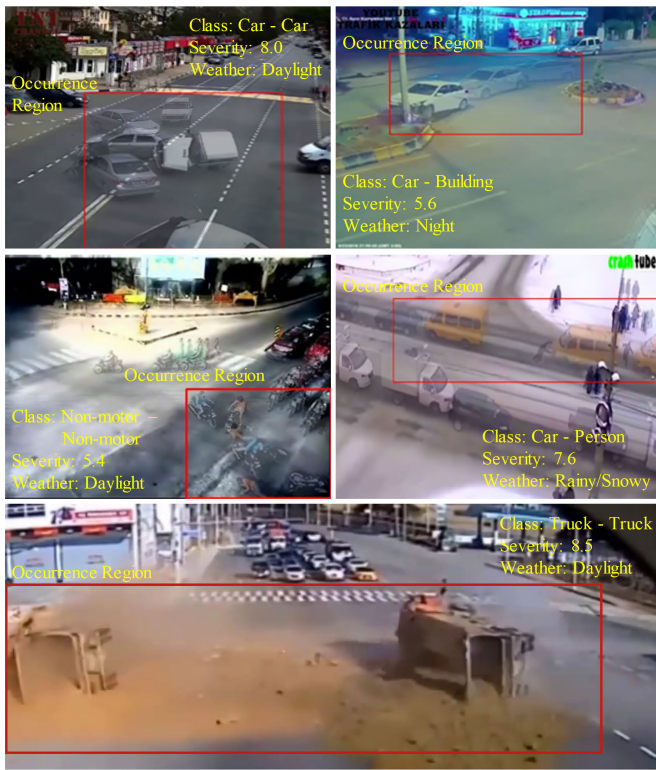


Fig. 1. Some representative examples of annotated frames in the FAD database for the tasks of multiple accident classification, temporal spatial occurrence region detection, and accident severity estimation. The four examples on the top and middle rows show the accident occurrence area covering the beginning to the end. The bottom one shows that the occurrence area contains not only the participants in the accident but also the background area affected. The accident category, severity score, and weather condition are shown in the corner of each example.

the overall situation of the accident, such as the severity of the accident and the affected area. Surveillance cameras can provide enough visual information for judgment. We novelly estimate the traffic accident severity while detecting the area for event-level analysis.

As the existing traffic accident databases only focus on 0/1 binary classification or temporal analysis, we collect and annotate a novel Fine-grained Accident Detection (FAD) Database that could support multiple tasks: temporal accident detection,

multiple accident classification, spatial accident region detection, and accident severity estimation. Some examples of annotated frames are shown in Figure 1.

With the development of deep learning technology, computer vision technology based on the convolutional neural network has developed rapidly. Deep learning computer vision technology does not require manual extraction of image/video features. The convolutional neural network could generate effective low-level features trained through a large amount of data. Therefore, computer vision technology based on deep learning has higher accuracy than traditional image processing methods based on manual features extraction. Recently, thanks to the strong representation capabilities of the transformer framework in the field of natural language processing, researchers are looking for ways to apply it to computer vision tasks, e.g., the transformer shows the effectiveness of local-global attention in the image object detection task [7], [8]. In this paper, we propose a Video 3D Transformer for Fine-grained Accident Detection for multiple accident classification, temporal-spatial accident detection, and accident severity estimation.

The main contributions of this paper are summarized below:

- We propose a novel Fine-grained Accident Detection (FAD) Database for more detailed accident analysis such as temporal accident detection, multiple accident classification, spatial accident region detection, and accident severity estimation.
- We propose a Video 3D Transformer for Fine-grained Accident Detection for multiple accident classification, temporal-spatial accident detection, and accident severity estimation.
- Various experiments are presented on the proposed FAD database, verifying the effectiveness of the proposed model and demonstrating that the FAD database is challenging.

II. RELATED WORK

A. Anomaly Detection

Traffic accidents are highly rare, heterogeneous, and occur randomly. Accidents are always considered a part of abnormal status in traffic scenarios. However, accident analysis remains

to distinguish between normality and abnormality. Anomaly detection has been studied for a long time, and the goal is to locate abnormal events in a video sequence in time or space. Due to the difficulty of modeling abnormal events, the abnormal status is defined as the most different from normal behaviors, such as bikers, small carts, and cars in people in UCSD Ped [12] and ShanghaiTech [13], wrong direction and no payment in Subway [14], running or throwing in public [15], [16], jaywalking or car illegally parked in the street [17], and crime in the real-world such as UCF Crime [11]. Whereas specific traffic accident detection could be defined in more detail than general anomaly detection.

The definition of anomaly is more akin to an unsupervised learning framework, which trains with only normal data and validates with both normal and abnormal data. In some contexts with temporary dependency, normal status is more predictable than abnormal status, [13] utilizes a video sequence with consecutive frames to generate a future frame as close as possible to the ground truth by adding intensity, gradient, and motion constraints. The constraints provide consistency in RGB space and minimize optical flow differences. Eric et al. [18] utilize the low-rank sparse representation of target videos with a corresponding similar decomposition performed on an anomaly-free reference video. Ye et al. [19] utilize an autoencoder GAN-based reconstruction framework, introducing a predictive coding module for refining predicted frames, which helps to distinguish the difference in anomaly scores between normal and abnormal frames. Meanwhile, [3] introduces a self-trained deep ordinal regression model to learn a ranking model that optimizes the anomaly score for unsupervised video anomaly detection.

Meanwhile, [11] formulates anomaly detection as a weakly supervised problem using multiple instance learning. Under this weak supervision, to distinguish between normal and abnormal frames, the loss function forces the maximum score of an abnormal video to be higher than that of a normal video. Zhong et al. [20] consider anomaly detection as a weakly supervised learning problem using noisy label learning, where the labeling of some frames in abnormal videos is incorrect. They trained a Graph Convolutional Network with more precise noise labels so that the classification network could be trained better with frame-level labels. Reference [21] introduces a high-order context encoding model that extracts semantic representations and dynamic variations focusing on temporally localizing anomalous segments within anomalous videos.

Semi-supervised learning is also used for anomaly detection, such as Abati et al. [22] jointly learning an autoencoder and a density estimation network with autoregressive network layers. And [23] combine a one-class SVM with neural networks, learning the one-class hyperplane from the low-dimensional representation by the neural network rather than the original input space. There are also some supervised methods based on temporally annotated training videos, such as [24], which introduce the object detector and multiple object tracking for generating the trajectories of anomaly vehicles. The highly compact feature set of a cell structure extracts the information of support regions in a coarse-to-fine fashion in [25].

Li et al. [26] utilize a variational U-Net by combining U-Net and conditional variational auto-encoders for the fusion of features at different scales.

For further fine-grained accident detection and analysis, the setting of supervised methods can meet the purpose of diversity, accuracy, and effectiveness.

B. Traffic Accident Database

Given the particularity of traffic accidents, a lot of research has been carried out specifically for traffic accident analysis, and some traffic accident databases have also been constructed. Two kinds of videos are used for traffic accident analysis: dashcam videos and surveillance camera videos. As shown in Table I, the dashcam video databases cover the anticipation (c), temporal detection (A3D [9]), and attention (DADA [10]) tasks which focus on the application for the owner of the dashcam such as personal cars. The surveillance camera video databases mostly cover temporal detection tasks (Iowa DOT [6], IITH [5], UCF Crime [11]) which focus on the accident event itself.

On both the dashcam and surveillance camera databases, the accident analysis mostly focuses on the 0/1 binary detection or prediction tasks. Only DADA [10] annotates the accidents by whether the owner of the dashcam is involved or uninvolved, into 62 accident categories in driving. However, DADA [10] just analyzes the driver's attention in different situations. For spatial traffic accident analysis, DAD [1], CCD [4], and CADP [2] provide the localizations of the objects, which could assist in analyzing traffic accidents by the relationship between objects. The object detectors (such as Faster R-CNN [27] and Cascade R-CNN [28]) are used to detect candidate objects compared with the annotated objects' position.

For further fine-grained accident detection and analysis, we construct a new database refining the definition of accident classification detection, temporal detection, and spatial detection.

III. FINE-GRAINED ACCIDENT DETECTION DATABASE

For traffic accident analysis, we construct a Fine-grained Accident Detection (FAD) Database for more detailed accident analysis.

Video Data: As the public databases have been used for traffic accident analysis, we collect most of the public accident video databases recorded by surveillance cameras, including CADP [2], Iowa DOT [6], IITH [5], and part of surveillance videos in UCF Crime [11] which contain traffic accidents, totaling 1,603 raw videos. Then we split the raw videos into 3,996 video clips by the normal videos and 26 fine-grained categories of whether the accident appears and which category the accident belongs to.

Annotations: Based on the collection of public surveillance accident video databases, we re-annotate the attributes and temporal-spatial definition of the accident. The specific annotation content is as follows:

- 1) Accident categories: We define the categories of the traffic accident by the participant in the accident, including hitting, scratching, and out of control. The participants are

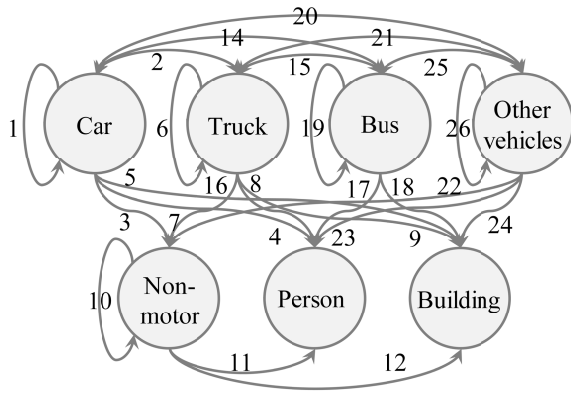


Fig. 2. The accident category graph. The connection represents the participants in each accident category.

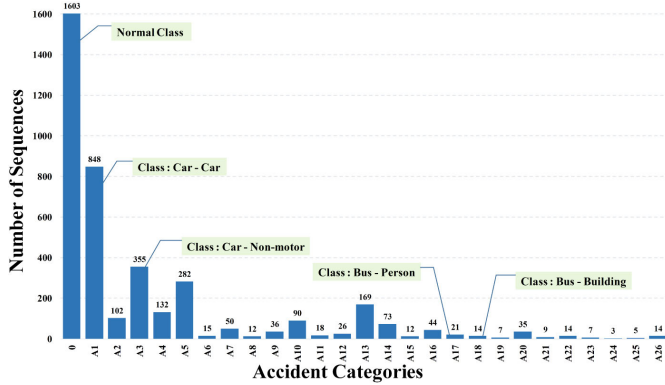


Fig. 3. The video sequence amount distribution w.r.t. accident categories in the FAD database. The abscissa represents the identity of each category. The labels of some representative categories are shown on each corresponding class.

grouped into seven classes including ‘Car’, ‘Truck’, ‘Bus’, ‘Non-motor’, ‘Person’, ‘Building’, and ‘Other vehicles’. The interaction diagram of the relationship between categories and participants is shown in Figure 2. For example, the class ‘A1’ is ‘Car - Car’ representing the accidents with the participants of ‘Car’ and ‘Car’. The subtle difference between class ‘A17’ and class ‘A18’ is the participant which interacts with ‘Bus’. In particular, we define the normal traffic videos as the class ‘0’ and the miscellaneous accidents excluding 25 categories of accidents in Figure 2 as the class ‘A13’. The quantitative statistics of each category are shown in Figure 3.

- 2) Start frame: the starting frame number of each accident. The starting frame is not the hitting moment but the accident beginning time at which the participants appear and could not avoid accidents.
- 3) End frame: the end frame number of each accident.
- 4) Occurrence region: a square area containing not only the participants’ position at the beginning and end of the accident but also the affected area by the accident.
- 5) Weather: weather condition of each accident. We define three categories of the weather condition, including: ‘Day-light’, ‘Night’, and ‘Rainy/Snowy’.
- 6) Severity score: the severity of the traffic accident contains: *Slight* (score 1 ~ 3), including scratches, failure, etc.

Medium (score 4 ~ 6) including minor collision, slow collision, etc. *Serious* (score 7 ~ 9) including body damage, vehicle deformation, fast collision with people/non-motor vehicles, building damage, etc.

Annotation items 1-5 are cross-checked for the accuracy of labeling content after a single round of labeling. Annotation item 6, i.e., the Severity score is the judgment of the subjective severity score of multiple people, and the original labeling content of multiple scores was recorded. The Severity score does not adhere to the fixed level of the traffic accident but is subjectively given according to the specific situation of the accident.

Compared with existing databases, our FAD database redefines the classification categories and covers multiple traffic accident analysis tasks: multiple accident classification, temporal-spatial accident region detection, and accident severity estimation which more emphasis places on the detection and analysis of the event itself, as shown in Table I.

IV. METHOD

For fine-grained traffic accident detection analysis, we construct a Video 3D Transformer to extract the video feature by combining the RGB feature and optical flow feature, which could better express the change in the scene, and introduce the multi-task loss for different fine-grained detection tasks.

A. Video 3D Transformer Architecture

The overall Video 3D Transformer architecture is depicted in Figure 4. We utilize the I3D [29] as the backbone feature to extract the RGB feature and optical flow feature. The backbone feature starts from the initial N_i RGB images $x_{rgb} \in \mathbb{R}^{N_i \times H_i \times W_i \times 3}$ (with 3 color channels) and corresponding optical flow input $x_{of} \in \mathbb{R}^{N_i \times H_i \times W_i \times 2}$ (with 2 channels) and generates a lower-resolution activation map $f_{rgb}, f_{of} \in \mathbb{R}^{N \times H \times W \times C}$.

1) *Encoder*: For the transformer encoder, we first construct an encoder-init block for fusing the RGB feature and optical flow feature. A $1 \times 1 \times 1$ convolution reduces the channel dimension of each high-level activation map f_{rgb} and corresponding optical flow f_{of} from C to a smaller dimension d , creating new feature maps $z_{img}, z_{of} \in \mathbb{R}^{N \times H_i \times W_i \times d}$.

The encoder expects a sequence as input, hence we utilize the multi-head self-attention module and feed-forward network (FFN) for both the RGB feature and optical flow feature. Then a multi-head module is introduced for combining the RGB feature and optical flow feature, in which the RGB feature is Value v , the optical flow feature with 3D position encoding is Key k , and the RGB feature with 3D position encoding is Query q . As the optical flow effectively expresses the changes in the scene where the accident would occur more probably, we utilize the optical flow feature as the Key in the multi-head attention module.

The 3D position encoding represents the position information of the sequence, which is defined as:

$$PE_{(n, pos, 2i)} = ((\sin(pos/10000^{2i/d})) + 2n)/N - 1 \quad (1)$$

$$PE_{(n, pos, 2i+1)} = ((\cos(pos/10000^{2i/d})) + 2n)/N - 1 \quad (2)$$

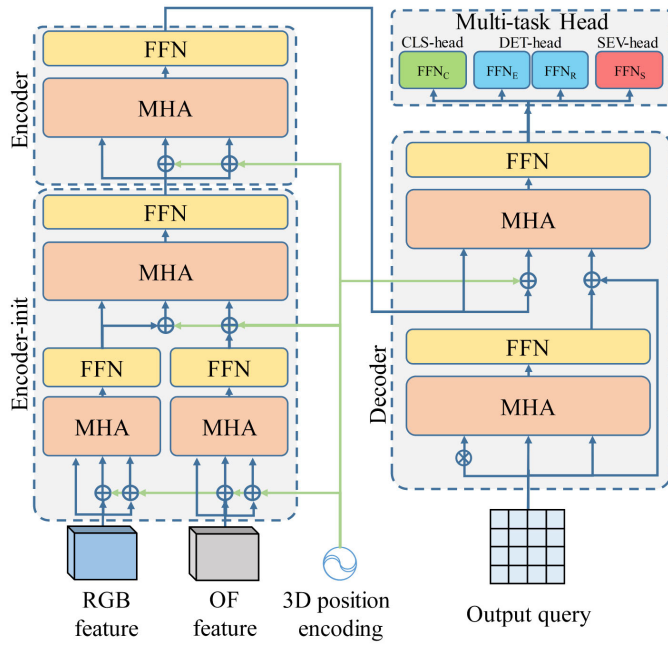


Fig. 4. The framework of the proposed model. The encoder-init block is constructed for fusing the RGB feature and optical flow feature with the 3D position encoding. The decoder generates the video features for the multi-task head inference using the encoder output feature with the spatial query. The CLS-head FFN_C infers the multiple accident classification. The DET-head FFN_E and FFN_R infer the accident existence and region offset at each position, and the SEV-head FFN_S infers the accident severity.

where n is the frame number in N frames of the feature, pos is the position in the n th frame feature, and i th dimension in d .

The RGB feature and optical flow feature would be fused as video feature $z_v \in \mathbb{R}^{N \times H_i \times W_i \times d}$ by the multi-head attention module and a feed-forward network followed. As the encoder layers expect a sequence as input, we collapse the spatial dimensions of z_v into one dimension, resulting in an $NHW \times d$ feature map. Each encoder layer followed has a standard architecture that contains a multi-head self-attention module and a feed-forward network.

2) *Decoder*: For the transformer decoder, we construct the decoder following the standard architecture of the transformer. As we expect to analyze the traffic accident in detail such as the spatial occurrence area detection, the position queries are defined as $m \times m$ which would correspond to the candidate detections and collapse into a $Nmm \times d$ sequence. The multi-head self-attention module and feed-forward network are also used for embedding the position queries to the encoder-decoder attention inputs. As the decoder would also be permutation invariant, the candidate inputs must be different. Thus, we introduce the same 3D position encoding to the input of each attention layer. Then in the multi-head self-attention module and feed-forward network that followed, we regard the encoder output feature as the Value v , the encoder output with 3D position encoding as Key k , and position embedding as Query q . The pair-wise temporal and spatial relations between the local and global information in the scene could be reasoned by using encoder-decoder attention.

The $m \times m$ position candidate queries with the RGB image and optical flow feature are transformed into an output embedding by the encoder and decoder. They are then independently inferred as multiple objectives in different tasks.

B. Multi-Task for Fine-Grained Accident Detection

The video feature is used for inferring different tasks (i.e., multiple accident classification and temporal-spatial accident occurrence region detection) which is produced by the transformer corresponding with the $m \times m$ position candidate queries.

1) *Multi-Task Head*: The final multiple inferences are computed by the head of each task. The video feature, i.e., the output of the decoder, is firstly extended in dimension, resulting in a $m \times m \times Nd$ feature map. The feature at each position is passed into densely connected layers for the final predictions. For multiple accident classification, one densely connected layer FFN_C using the whole video feature predicts the class label followed by a softmax function, as shown in Figure 4. For temporal detection and spatial region detection, one densely connected layer FFN_E predicts the determination of whether an accident exists and the center-ness of the region, and one densely connected layer FFN_R predicts the normalized center coordinates offset, height, and width of the region box. For accident severity estimation, one densely connected layer FFN_S using the whole video feature predicts the severity of the accident video.

2) *Multi-Task Loss*: We utilize the multiple losses for each detection task. The loss function of our multi-task network is formulated as

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{bbox}} + \mathcal{L}_{0/1} + \mathcal{L}_{\text{sev}} + \|w\|_2^2. \quad (3)$$

The loss function contains four parts: the multiple classification loss $\mathcal{L}_{\text{cls}}$, the spatial region detection loss $\mathcal{L}_{\text{bbox}}$, the discrimination loss $\mathcal{L}_{0/1}$ and the severity estimation loss $\mathcal{L}_{\text{sev}}$. The term $\|w\|_2^2$ is the L2 regularization constraint for parameters w of the network.

The classification loss $\mathcal{L}_{\text{cls}}$ is a focal loss, which is generated by the result of multiple accident classification and is defined as:

$$\mathcal{L}_{\text{cls}} = \sum_{i=0}^C -\alpha(1 - p_i)^\gamma c_i \log(p_i), \quad (4)$$

where c_i is the ground truth category label of video clip classification. p_i is the probability of each class. α is the factor controlling the shared weight of each class sample with the cross-entropy loss. γ is the modulating factor reducing the weight of the samples that are easy to classify and making the model to focus more on the samples that are difficult to classify during training.

The spatial region detection loss $\mathcal{L}_{\text{bbox}}$ consists of bounding boxes IoU loss $\mathcal{L}_{\text{IoU}}$ and center-ness loss $\mathcal{L}_{\text{ctr}}$ which are defined as:

$$\mathcal{L}_{\text{bbox}} = \mathcal{L}_{\text{IoU}} + \mathcal{L}_{\text{ctr}}, \quad (5)$$

$$\mathcal{L}_{\text{IoU}} = -g_1 \log(\text{IoU}(b_p, b_g) + \epsilon), \quad (6)$$

$$\mathcal{L}_{\text{ctr}} = -g_{\tau} \log \frac{1}{1 + e^{-\tau}}, \quad (7)$$

where b_p and b_g are the predicted region and ground truth of the accident region respectively. $\text{IoU}(b_p, b_g)$ is the Intersection over Union of the two bounding boxes. g_1 is the label of the positive class. ε is a minuteness item for preventing undesired zero. $\mathcal{L}_{\text{ctr}}$ is the center-ness loss which is the sigmoid binary cross entropy loss between the ground truth g_{τ} of center-ness at each location with the region bounding box and predicted center-ness output τ by the DET-head at the corresponding location. The center-ness of a bounding box at one location is defined similarly as [30], $\text{centerness} = \sqrt{\frac{\min(l,r)}{\max(l,r)} \times \frac{\min(t,b)}{\max(t,b)}}$, computing the regression offset targets l, t, r, b at the location estimating the distance from the target center.

The determination loss $\mathcal{L}_{0/1}$ is a soft-max cross-entropy loss, which is generated by the result of whether an accident exists at one location and is defined as:

$$\mathcal{L}_{0/1} = \frac{1}{m^2} \sum -(g_0 \log q_0 + g_1 \log q_1), \quad (8)$$

where g_0 and g_1 are the ground truth labels of the determination class at each location of $m \times m$ position candidate queries. p_0 and p_1 are the probabilities of the negative or the positive class at the corresponding location.

The severity estimation loss $\mathcal{L}_{\text{sev}}$ is given by the smooth-L1 loss function, defined as:

$$\mathcal{L}_{\text{sev}} = \begin{cases} 0.5(s_p - s_g)^2 & \text{if } |s_p - s_g| < 1 \\ |s_p - s_g| - 0.5 & \text{otherwise} \end{cases} \quad (9)$$

where $|s_p - s_g|$ is the difference between the predicted severity score s_p and the ground-truth severity score s_g .

V. EXPERIMENTS

In this section, we evaluate our model on the proposed FAD database with various tasks, including multiple accident classification and temporal-spatial accident occurrence region detection. As accident detection requires robust models due to accident contingency, we want to test on enough samples and thus split it half and half for the training and testing set, respectively. The FAD database would be made available later including the repartitioning of data and the fine-grained annotation of each detection task.

A. Evaluation Metrics

For multiple accident classification, we utilize the Average Precision (AP) to evaluate the correctness of identifying an accident category from a video. We consider not only the total categories but also the 0/1 classification (AP#2) and the specific accident classification (AP#26). We also introduce the Receiver Operating Characteristic and Area Under Curve (AUC) for evaluating the methods.

For temporal-spatial accident occurrence region detection, we use mean Average Precision respectively at IoU=0.33, 0.5, 0.66, 0.75 (AP₃₃, AP₅₀, AP₆₆, AP₇₅) to assess the accuracy of the methods. If IoU is larger than a threshold, then the region prediction is positive to contain an accident, otherwise, it is negative.

For severity estimation, we use the Mean Squared Error (MSE), Mean Absolute Error (MAE), Pearson Correlation Coefficient (PCC), and Spearman's Correlation Coefficient (SPC). As the severity estimation score is a relatively subjective judgment from multiple people, different from the accurate classification or temporal-spatial detection, we use the geometric mean score of the annotation scores by multiple people for unifying scores and reducing the effect of outliers.

B. Implementation Details

We implement our model with TensorFlow. The original I3D framework [29] is introduced for generating the RGB and optical flow feature as the backbone. For the input, we evenly select 16 frames of each video clip as the initial N_i images and resize each frame to 224×224 . As the accident occurrence region is relatively large, we choose the output of the 'Mixed_5c' block from the feature maps of the backbone as the RGB and optical flow feature maps, the size of which is $2 \times 7 \times 7$. For the proposed video 3D transformer, except for the encoder-init block, we utilize 2 block encoder layers and 2 block decoder layers respectively. We set 512 for the hidden feature dimension in both the encoder and decoder of the transformer. For a fair comparison with the baseline I3D method, position candidate queries of the decoder are set to 7×7 with the same width and height as the feature map of the 'Mixed_5c' block. For the loss parameters, we set the α to 0.25, γ to 2, and ε to 1e-5.

To compare with baseline models, we combine the multi-task head and the I3D feature maps as the baseline models in which the RGB features and optical flow features are concatenate-joined (I3Dc) and mean-joined (I3Dm) respectively, while only the original RGB features are used (I3D). For a more comprehensive view of the proposed method's performance, we also utilize the C3D [31] feature maps as the baseline models, correspondingly in which the RGB features and optical flow features are concatenate-joined (C3Dc) and mean-joined (C3Dm) respectively, while only the original RGB features are used (C3D). We also compare the common abnormal detection method VABD [26] on the proposed FAD database. We train the baseline models and our model both by setting the initial learning rate to 10^{-4} for 20 epochs, 10^{-5} for 40 epochs and 10^{-6} for 40 epochs as a total of 100 epochs with the batch size set to 32 on 16 V100 GPUs.

C. Multiple Accident Classification

A quantitative comparison between our proposed model (FADTrans_C based on the C3D features and FADTrans_I based on the I3D features) and the baseline models is shown in Figure 5. We present classical precision-recall curves on the testing set of the FAD database. The AP is computed on all the categories that contain the normal traffic class '0' and 26 accident categories. From the figure, we could see that our models with a 3D Transformer based on two kinds of features achieve much better accuracy on AP than the baselines. Compared with the C3D features, I3D features could offer more effective video features for accident analysis. Given the imbalance of samples in the categories in the proposed

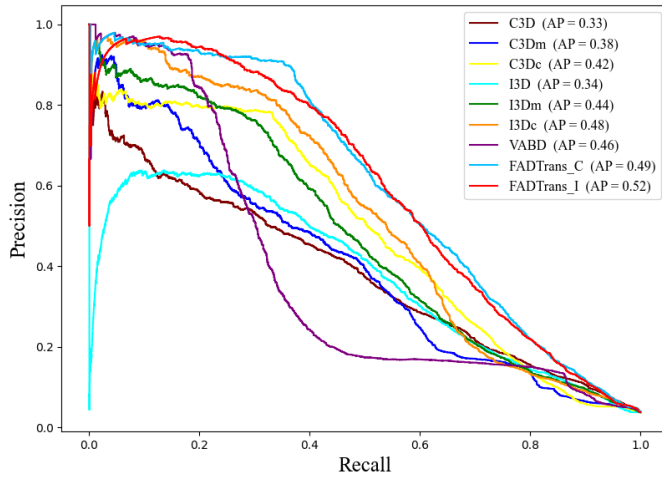


Fig. 5. Precision-Recall plot for the testing set of the FAD database. The legend presents the AP score of each method.

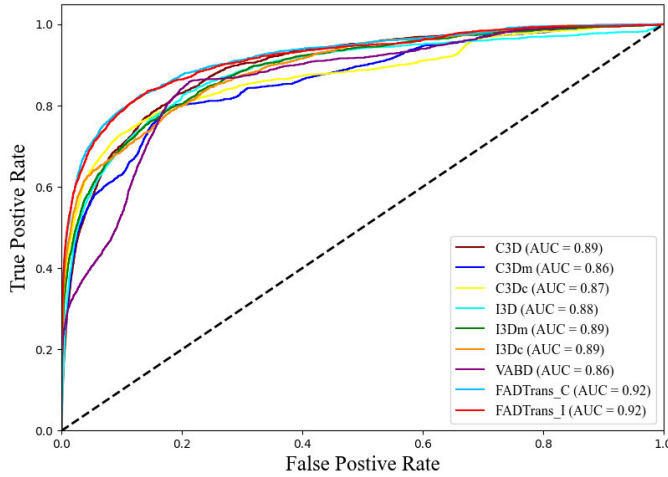


Fig. 6. ROC curve for the testing set of the FAD database. The legend presents the AUC score of each method.

database, the receiver operating characteristic (ROC) curves on the testing set are shown in Figure 6. The curves show that the result of our method is much smoother and gets a higher AUC value, which means our model could be much more robust and accurate. The original I3D with only RGB features shows relatively poor performance without the optical flow information. Meanwhile, the I3Dc model, which concatenate-joins the RGB feature and optical flow feature, achieves better AP than the mean-joined model I3Dm, while the ROC curves of them show close performance. This divergent result may occur because the concatenate-joined feature keeps more dimension information and the simple feature fusions achieve similar reliability performance. To verify the effectiveness of the proposed method as well as the rationality of the proposed FAD dataset, we compare the AP and AUC on the accident testing set of the UCF Crime database, as shown in Table II. From the table, it could be seen that the methods achieve relatively good precision on the traditional accident analysis database which only detects the existence of the accident. The methods C3Dc and I3Dc using features concatenate-joining both RGB features and optical flow features could

TABLE II
COMPARISON WITH THE METHODS ON VARIOUS AVERAGE PRECISION AND AREA UNDER CURVE ON THE ACCIDENT TESTING SET OF THE UCF CRIME DATABASE

	AP	AUC
C3D [31]	72.09%	76.32%
C3Dm [31]	70.61%	76.84%
C3Dc [31]	87.65%	86.43%
I3D [29]	79.20%	79.24%
I3Dm [29]	84.05%	83.44%
I3Dc [29]	90.09%	85.65%
VABD [26]	93.53%	87.92%
FADTrans_C	96.23%	89.23%
FADTrans_I	96.83%	89.47%

TABLE III
COMPARISON WITH THE METHODS ON VARIOUS AVERAGE PRECISION ON THE TESTING SET OF THE FAD DATABASE. AP INDICATES THE AVERAGE PRECISION OVER ALL CLASSES. AP#26 INDICATES THE AVERAGE PRECISION OVER THE SPECIFIC ACCIDENT CLASSES. AP#2 INDICATES THE RESULTS OF BINARY 0/1 CLASSIFICATION REGARDING ALL THE ACCIDENTS AS THE POSITIVE SAMPLE

	AP	AP#26	AP#2
C3D [31]	33.14%	16.18%	77.15%
C3Dm [31]	37.64%	16.74%	83.01%
C3Dc [31]	42.26%	19.49%	84.63%
I3D [29]	33.72%	18.28%	82.98%
I3Dm [29]	43.83%	20.35%	92.81%
I3Dc [29]	48.05%	22.47%	95.46%
VABD [26]	45.75%	20.03%	95.48%
FADTrans_C	48.87%	20.50%	95.18%
FADTrans_I	51.84%	26.16%	97.49%

get much better results, while our method achieves excellent performance with 96.83% on AP and 89.47% on AUC using two types of features more effectively. By contrast, the APs of results significantly decrease on the proposed FAD database for multiple tasks that is much more challenging.

We also compare our proposed model with the baseline model on AP of only 26 accident categories and binary 0/1 classification, as shown in Table III. For conventional binary 0/1 classification, we consider all kinds of traffic accidents as the positive category and normal traffic videos as the negative ones. Both the proposed model and the baseline model achieve excellent performance on binary classification, especially since our model gets 97.49% on AP#2. This result may occur because there are obvious differences in scene modes between normal traffic videos and traffic accident videos. The common abnormal detection method VABD [26] also performs well with 95.48% on AP#2. However, the performance on the AP#26 is much worse, which only evaluates the accuracy on the 26 accident categories i.e., class A1~A26. Our model FADTrans_I gets only 26.16% on AP#26, even VABD with 20.03%. The reason for this decrease in performance is due to the variety of accidents. Even with the refined definition of the accident category, the fine-grained accident classification is quite challenging.

The APs of each category are shown in Figure 7. From the figure, we could see that the models all get outstanding performance on normal traffic videos class '0'. The distribution of high and low performance on specific accident categories is roughly similar to the distribution of data quantity in the

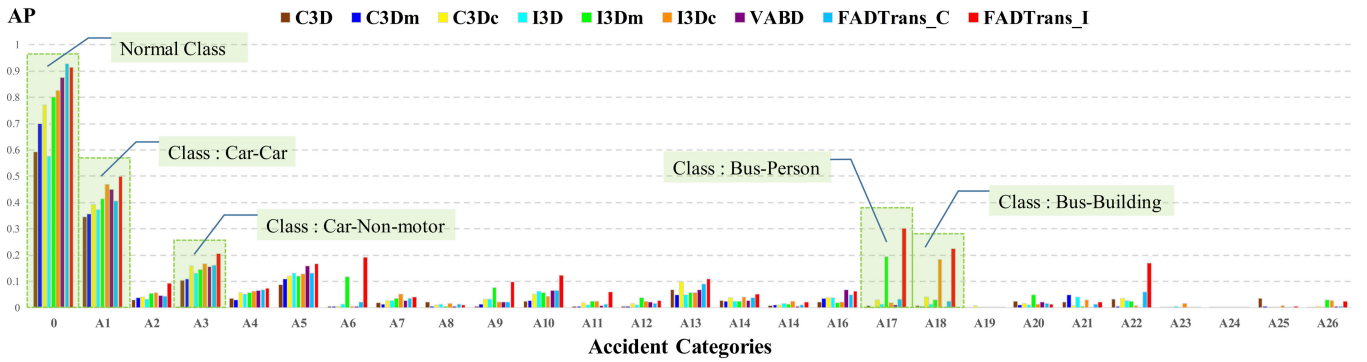


Fig. 7. Results of a quantitative comparison of the methods on each accident category. The abscissa represents the identity of each category. The labels of some representative categories are shown on each corresponding class.

TABLE IV

COMPARISON WITH THE METHODS ON THE AVERAGE PRECISION ACCORDING TO WEATHER ATTRIBUTE ON THE TESTING SET OF THE FAD DATABASE

	Daylight	Night	Rainy/Snowy
C3D [31]	32.11%	34.08%	38.13%
C3Dm [31]	37.01%	38.25%	42.35%
C3Dc [31]	42.27%	46.51%	44.43%
I3D [29]	32.95%	36.15%	37.64%
I3Dm [29]	42.64%	46.19%	49.31%
I3Dc [29]	47.47%	49.44%	51.49%
VABD [26]	44.18%	48.10%	50.15%
FADTrans_C	48.13%	51.26%	53.77%
FADTrans_I	51.33%	53.52%	55.99%

TABLE V

COMPARISON WITH THE METHODS ON VARIOUS AVERAGE PRECISION ON THE TESTING SET OF THE FAD DATABASE. AP₇₅, AP₆₆, AP₅₀, AND AP₃₃ INDICATE AVERAGE PRECISION AT IoU=0.75, 0.66, 0.5, 0.33, RESPECTIVELY

	AP ₇₅	AP ₆₆	AP ₅₀	AP ₃₃
C3D [31]	0.16%	1.54%	9.87%	36.65%
C3Dm [31]	0.57%	2.18%	11.16%	36.41%
C3Dc [31]	0.51%	2.19%	12.38%	38.11%
I3D [29]	0.24%	1.78%	10.59%	31.87%
I3Dm [29]	0.49%	2.27%	11.49%	38.99%
I3Dc [29]	0.81%	2.59%	15.97%	42.96%
VABD [26]	2.05%	3.72%	19.9%	47.81%
FADTrans_C	2.02%	6.39%	24.92%	50.02%
FADTrans_I	2.43%	8.17%	28.24%	51.94%

proposed database. This indicates that increasing the sample number of each fine-grained category could effectively learn the occurrence mode of accidents discriminatively. The results are divergent in some categories with small samples such as class A17 and A18. The reason may be that the participants in class A17 and A18 contain the ‘Bus’ which has obvious appearance specificity. On the contrary, the results of class A13 with more samples are unsatisfactory due to the miscellaneous accidents.

To further explore the performance of the models in various scenarios, in Table IV, we evaluate these models’ performance on different weather attributes. From the table, the performance is generally worse in *Daylight* than *Night* and *Rainy/Snowy*. The reason for this decrease in performance is that even if the objects are clearer in daylight than at night, the background is more cluttered in daylight and the association between objects is more complex. The accidents in *Night* are distinguished more easily with patterns influenced by light and darkness, while rain or snow makes the background more concise resulting in better performance in *Rainy/Snowy*.

D. Temporal Spatial Occurrence Region Detection

We compare the proposed model with the baseline models on the testing set of the FAD database for the occurrence region detection task. The results of AP at different IoU thresholds over which the region detection is considered positive are shown in Table V.

Compared with the baseline models, our model achieves better performance on the AP at various IoU thresholds.

However, from the table, it could be seen that the result of occurrence region detection is not satisfactory as the best AP₇₅ is only 2.43% by our model. And our model achieves the best 28.24% on AP₅₀ while the performance is much higher on the same metric in other object detection tasks e.g., DETR [7] with 62.4 on AP₅₀. This shows that the occurrence region detection at the event level is very challenging and the whole process of the accident needs to be considered. The region detection by the I3Dc model is better than the I3Dm model, similar to the situation between C3Dc and C3Dm, due to the concatenate-joined feature providing more dimension information. With the effective motion consistency feature of the scene, the VABD gets 24.92% on AP₅₀ by detecting the abnormal location. Comparing the result of 2.43% on AP₇₅ and 51.94% on AP₃₃, the model could roughly locate the location of the accident, but it is difficult to accurately detect the occurrence region.

We also evaluate the performance of the models on different weather attributes, and the results are compared on the AP₅₀ as shown in Table VI. The performance of each model is similar in *Daylight* and *Rainy/Snowy* and decreases in *Night*. This divergent result may occur because the appearance feature is effective for the detection with the clear scene in *Daylight* and *Rainy/Snowy* and loses the effect due to the darkness in *Night*.

To further illustrate the ability of the proposed model, we visualize some occurrence region detection results on the testing set of the FAD database. For a better visual effect, we choose the results of the best-performing model versions of each method for comparison. As shown in Figure 8, even if our model could preferably learn to locate the occurrence

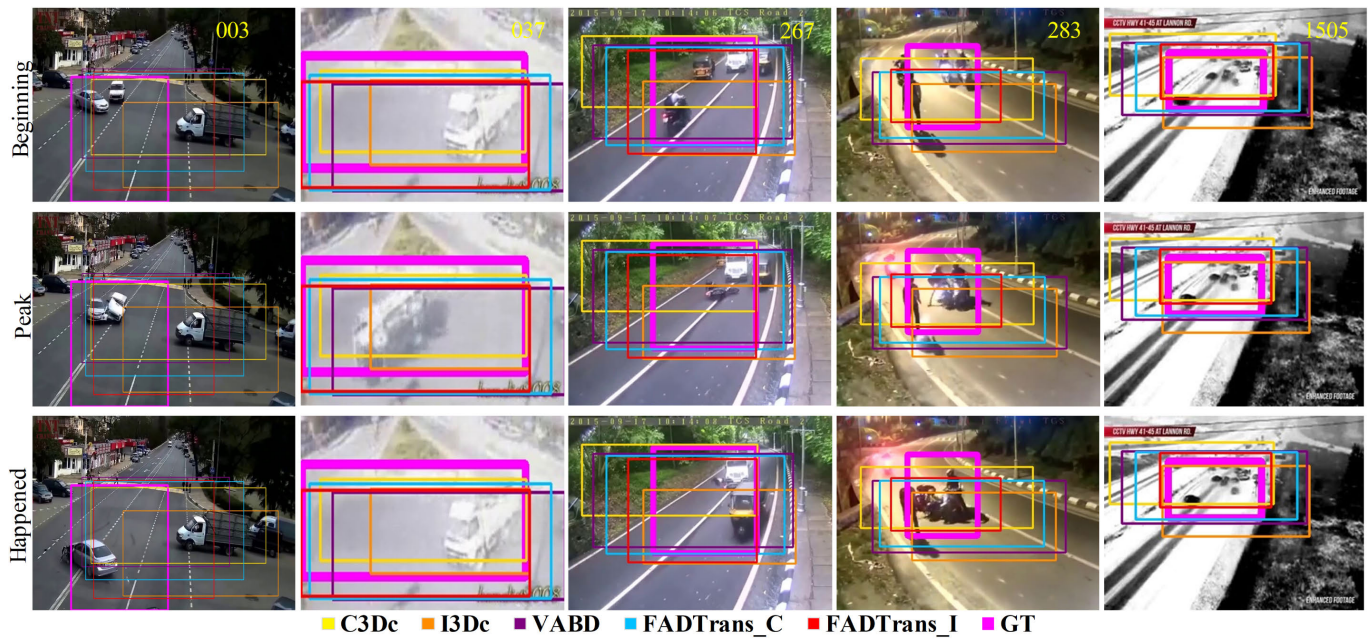


Fig. 8. Visualized comparisons of occurrence region detection on the testing set of the FAD database. The sequence numbers are shown at the top. Each row shows the beginning, peak, and happened state of the accident respectively. The magenta bold bounding boxes indicate the ground truth. The colored boxes indicate the results of the corresponding methods respectively. Our method can estimate a more accurate accident occurrence region.

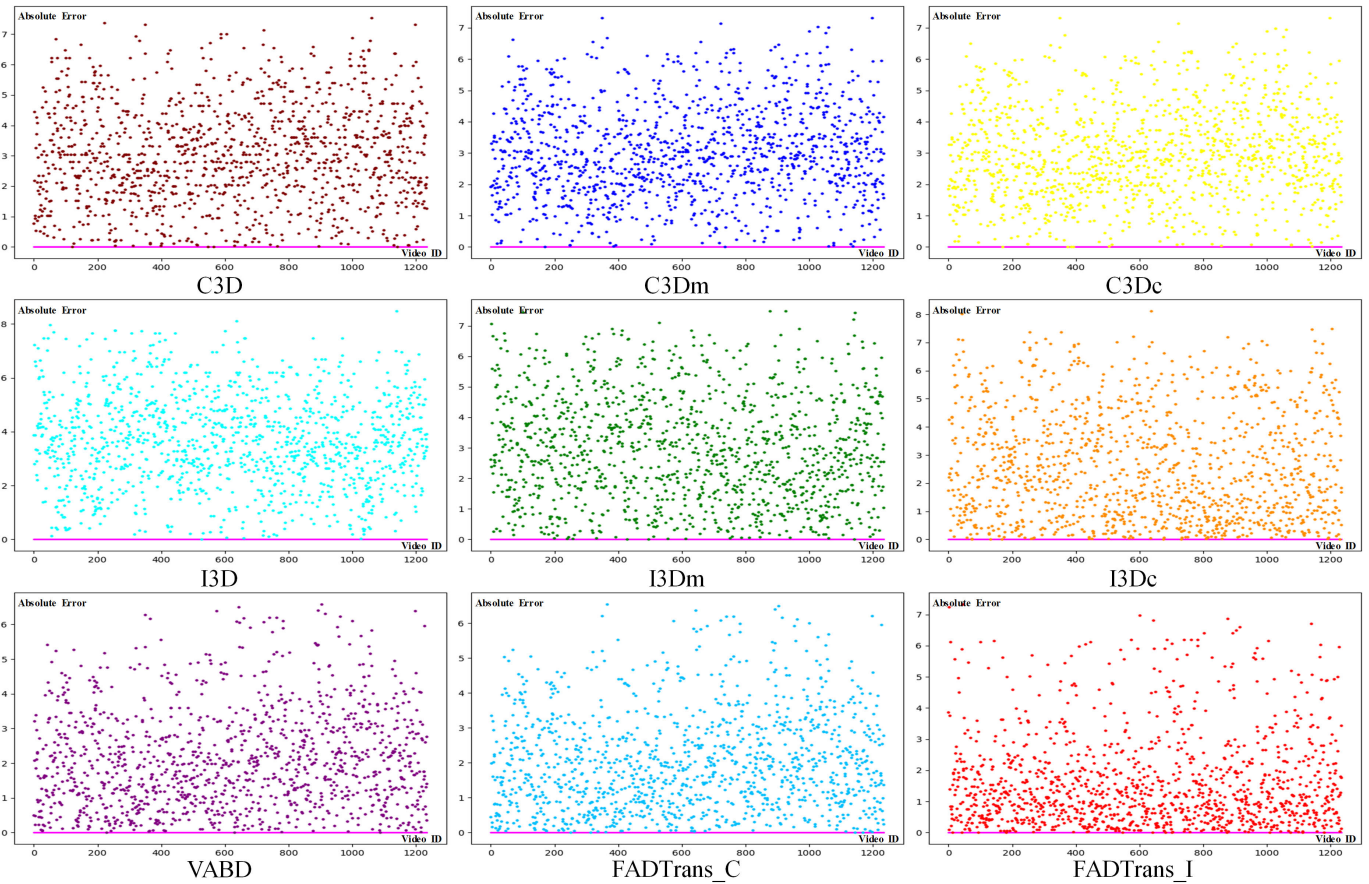


Fig. 9. Absolute Error for the testing set of the FAD database. The abscissa represents the identity of the test video. The points that are close to the magenta line mean that the results of accident severity estimation are more accurate.

region, the bounding box of the region could not fit tightly to the real occurrence area. It suggests that the occurrence region detection of the accident in videos still needs to be studied emphatically.

TABLE VI

COMPARISON WITH THE METHODS ON AVERAGE PRECISION ACCORDING TO WEATHER ATTRIBUTE ON THE TESTING SET OF THE FAD DATABASE. THE VALUES ARE THE AVERAGE PRECISION AT IOU=0.5

	Daylight	Night	Rainy/Snowy
C3D [31]	9.91%	9.96%	9.60%
C3Dm [31]	11.28%	12.99%	8.59%
C3Dc [31]	12.64%	14.72%	8.58%
I3D [29]	11.15%	9.09%	9.59%
I3Dm [29]	12.76%	9.59%	10.38%
I3Dc [29]	16.73%	12.63%	16.02%
VABD [26]	21.56%	16.45%	17.16%
FADTrans_C	26.02%	22.08%	23.74%
FADTrans_I	29.12%	24.24%	29.29%

TABLE VII

COMPARISON WITH THE METHODS ON MSE, MAE, PCC, AND SPC ON THE TESTING SET OF THE FAD DATABASE

	MSE	MAE	PCC	SPC
C3D [31]	11.75	2.36	0.55	0.49
C3Dm [31]	7.39	1.91	0.67	0.71
C3Dc [31]	7.12	1.86	0.67	0.71
I3D [29]	11.16	2.34	0.53	0.44
I3Dm [29]	7.24	1.81	0.68	0.71
I3Dc [29]	6.37	1.58	0.64	0.75
VABD [26]	4.93	1.29	0.79	0.78
FADTrans_C	3.71	1.23	0.79	0.76
FADTrans_I	3.01	1.01	0.82	0.77

E. Accident Severity Estimation

We compare the proposed model with the baseline models on the testing set of the FAD database for the severity estimation task. The results of various metrics are shown in Table VII.

Compared with the baseline models, our model achieves much lower error performance on the MSE and MAE. It could be seen that the fusion of the RGB features and optical flow features by the proposed model could offer more effective information about the accident event than the original RGB features or simple fusion. Effective scene consistency modeling could lead to more accurate severity estimation. The correlation coefficients between the ground truth severity score and the estimated result from the proposed model are much better than the other baselines. While the severity scores of the accident with various levels are subjectively judged by multiple people, our model achieves better-ranked severity estimation from the SPC results.

To further illustrate the ability of the proposed model, we visualize the absolute error results on the testing set of the FAD database. As shown in Figure 9, the proposed model estimates much closer severity scores to the ground truth. Meanwhile, there are still many severity estimation scores that are far from the ground truth, illustrating that the severity estimation task is quite challenging.

VI. CONCLUSION

In this paper, we presented a challenging FAD database in traffic surveillance scenarios. We define much more detailed accident analysis tasks, i.e., temporal accident detection, multiple accident classification, spatial occurrence region detection,

and severity estimation. For fine-grained accident detection, we proposed a novel method based on a Video 3D Transformer. We also present the results of our model and other baseline models on our database, which show the effectiveness of our model for fusing the video features on each task. In addition, the experimental results showed that fine-grained accident detection is more challenging than classical accident analysis such as binary classification or object detection. In the future, since the accident is related to the temporal-spatial context, the scene priors should be considered in fine-grained accident analysis.

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